

Project Development Phase

Model Performance Test

Date	10 February 2025
Team ID	SWTID-2026-2051
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot									
1.	Metrics	This project is a role-based access control (ACL) implementation, not a trained ML model. The metrics below are the access-decision test results for the READ ACL, treating "record visible to a bb1 user" as the positive class (6 records: 2 EEE, 4 ECE/CSE). Regression Model: MAE, MSE, RMSE, R2 score – Not applicable (no numeric prediction). Classification Model: Confusion Matrix (bb1 user): <table><tr><td></td><td>Pred: Visible</td><td>Pred: Hidden</td></tr><tr><td>Actual: EEE</td><td>2 (TP)</td><td>0 (FN)</td></tr><tr><td>Actual: ECE/CSE</td><td>0 (FP)</td><td>4 (TN)</td></tr></table> Accuracy Score: 6 / 6 = 100% Classification Report: Visible class – precision 1.00, recall 1.00, F1 1.00 (support 2). Hidden class – precision 1.00, recall 1.00, F1 1.00 (support 4). Other users tested: No-role user sees 0 of 6 records (expected 0); admin sees 6 of 6 (expected 6).		Pred: Visible	Pred: Hidden	Actual: EEE	2 (TP)	0 (FN)	Actual: ECE/CSE	0 (FP)	4 (TN)	EEE-only list for the bb1 user, empty list for the user without the role, and the admin full list.
	Pred: Visible	Pred: Hidden										
Actual: EEE	2 (TP)	0 (FN)										
Actual: ECE/CSE	0 (FP)	4 (TN)										
2.	Tune the Model	Hyperparameter Tuning: Not applicable. The equivalent tuning step was refining the READ ACL script from a basic role check to a layered check: (1) admin gets full access, (2) bb1 gets access limited by the data condition Branch is EEE, (3) everyone else is denied.	Final ACL script (Advanced Condition) and the delete confirmation screen.									

S.No.	Parameter	Values	Screenshot
		Validation Method: Impersonation testing. Impersonate the EEE user (bb1), a user without the role, and the admin, open u_institution_details.list, and compare the visible records with the expected result. Create, write and delete ACLs were validated the same way with bb1+bb2, bb1+bb2+bb3 and bb1-bb4 roles. Result after tuning: Validation accuracy 100%.	